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Comparison of Electromyographic Activity During Hip Extension Exercises Under Gravitational or Inertial Loading Conditions

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Background: Hamstring injury prevention programs include strengthening, especially eccentric exercises using both gravitational and inertial loading. Inertial exercises are characterized by eccentric contractions of high intensity and velocity. This study aimed to analyze the muscular activation of the biceps femoris (BF), semitendinosus (ST), gluteus maximus (GM), and gracilis (GC) muscles during hip extension (HE) exercises performed under both gravitational and inertial loading conditions.

Hypothesis: Inertial training would generate a greater activation of HE muscles than gravitational training.

Study Design: Cross-sectional study.

Level of Evidence: Level 4.

Methods: Fifteen resistance-trained men performed the unilateral straight knee bridge (SKB), 45° of HE, and stiff-leg deadlift (SDL) exercises under gravitational and inertial loading conditions. Concentric and eccentric phases were identified with a linear encoder. Differences between load types, exercises, and their interaction were examined to establish the electromyographic (EMG) activity of each muscle and BF/ST ratio.

Results: In the concentric phase, inertial loading showed a higher normalized EMG than gravitational loading for BF, ST, and GM. SKB and HE activated BF and ST between 9.6% and 24.3% more than SDL. In the eccentric phase, the inertial modality achieved greater GM activation than the gravitational form (18.1%). BF activation was increased with HE and SKB as compared with SDL (24.4% and 16.4%, respectively), while ST activation was likewise enhanced with HE as compared with SDL (15.1%).

Conclusion: Inertial training is more effective than gravitational training for the concentric activation of the hamstring muscles while SDL showed lower hamstring activation than HE and SKB. Therefore, HE and SKB with inertial loading should be taken into account in hamstring training programs.

Clinical Relevance: Inertial training is more effective than gravitational training for the concentric activation of the hamstring muscles. HE and SKB with inertial loading should be taken into account in hamstring training programs.

Keywords: muscle power; iso-inertial; resistance training; physiology

Hamstring strains are among the most frequent injuries in soccer, Australian football, rugby, and track sprinters, accounting for between 12% and 26% of all reported injuries.[7,8,16,27] Furthermore, hamstring injuries cause an important reduction in sports performance and loss of participation in running-based sports due to the high rates of injury recurrence and persistence of symptoms.[25]

While multiple risk factors for injury have been described, including unalterable factors (eg, age, previous injury, and ethnicity) and modifiable factors (eg, strength imbalances, flexibility, and fatigue),[25] the main mechanism of injury may be related to poor eccentric hamstring control during the late swing phase of running.[25] This is because during this phase, hamstring muscles function eccentrically to resist hip flexion

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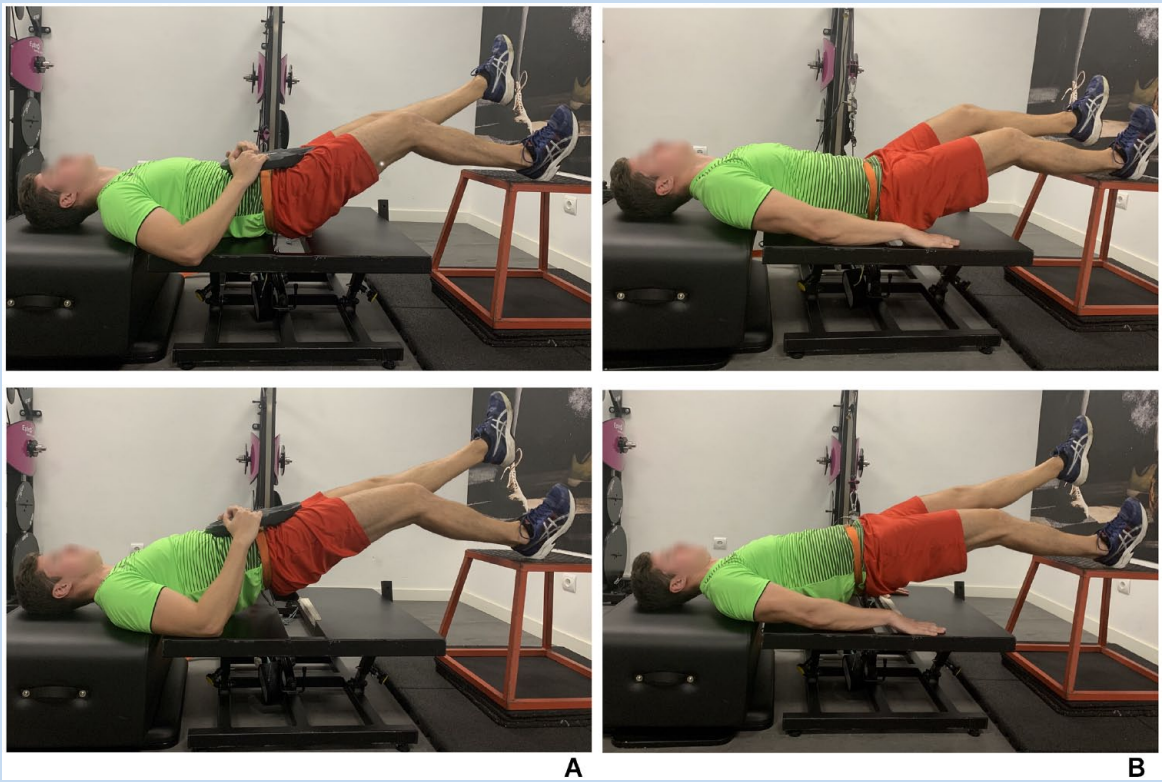

Figure 1. Unilateral straight knee bridge: gravitational (A) and inertial (B).

and decelerate knee extension,[18,30] absorbing energy from the swing limb before foot contact.[16] In particular, the biceps femoris (BF) undergoes the greatest change in length and performs the greatest amount of negative work during this time,[9] possibly explaining why the BF represents 80% of all hamstring injuries.[11]

Injury prevention programs include hamstring strengthening, especially eccentric exercises using both gravitational[13,14] and inertial loading.[3,12] The Nordic hamstring exercise is the most commonly used gravitational exercise in prevention programs,[15] but it is characterized by eccentric flexing of the knee, preferably activating the semitendinosus (ST) muscle.[5] Hence, recent proposals have suggested alternatives to the Nordic hamstring exercise that focus on extending the hip to selectively activate BF long head muscle, such as 45° of hip extension (HE).[6]

Inertial exercises are characterized by eccentric contractions of high intensity and velocity,[22] with the leg curl being the inertial exercise commonly used to prevent hamstring injuries.[3,12] This exercise reduces injury rates,[3,12] but has not yet been compared with other inertial hamstring exercises. On the other hand, inertial HE exercises have not been analyzed in terms of muscle activation to determine if the activation patterns in gravitational exercises are consistent with those of inertial training. This analysis would be necessary before implementation in hamstring prevention programs.

Therefore, the main objective of our study was to analyze the muscular activation of the BF and ST during HE exercises performed both gravitationally and inertially and to examine differences. Furthermore, a secondary objective was to record the electromyographic (EMG) activity of the gluteus maximus (GM) and gracilis (GC) muscles to identify the amount of additional activity associated with the activity of the hamstring muscles.

METHODS

Study Design

A within-subjects, repeated-measures design was used to analyze the surface EMG activation of the BF, ST, GM, and GC during 3 HE exercises, performed under gravitational and inertial loading conditions. Unilateral straight knee bridge (SKB), HE, and stiff-leg deadlift (SDL) (Figures 1-3) were chosen as these exercises have been shown to achieve increased BF activation in relation to the ST.[6] To compare the EMG at similar intensities between different types of exercises, the power in the concentric phase was the training variable selected. All participants completed 3 sessions spaced 1 week apart, supervised by a physical therapist experienced in the use of strength training with inertial devices for injury prevention. The first session was intended for participants to become familiarized with the exercises, while in the second, the

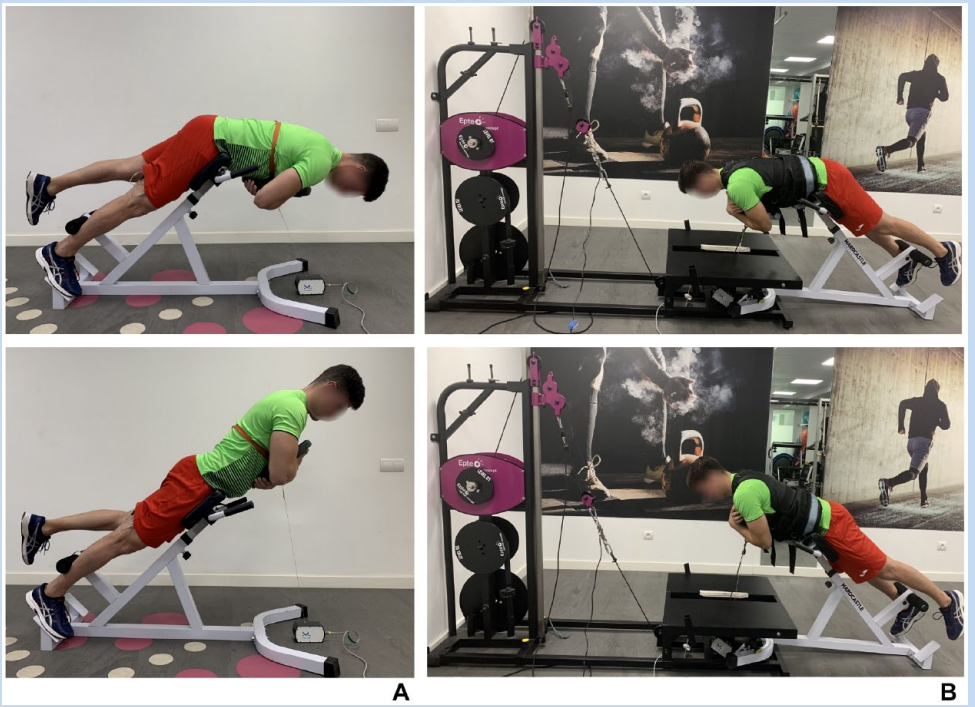

Figure 2. Forty-five degrees of hip extension: gravitational (A) and inertial (B).

maximum power was obtained for each exercise. The testing order was randomized and counterbalanced. Finally, in a third session, the EMG was obtained using the gravitational or inertial load for each exercise of the second session. The experimental protocol was approved by the Ethics Committee of the University of Valencia (Spain) (H1551979435533).

Participants

Fifteen men (mean age, 22.4 ± 2.5 years; body mass, 77.2 ± 9.8 kg; stature, 179.5 ± 7.4 cm; weekly physical activity, 434.0 ± 169.2 minutes; strength training experience, 4.3 ± 2.2 years) who were resistance trained (eg, recreationally trained athletes and powerlifters) but flywheel-naive were recruited by email via the University of Valencia intranet. Each subject had at least 2 years of resistance training experience and performed no resistance training exercises targeting the lower extremities in the 72 hours before the testing session. They should not have any injury, disease, or pain that could reduce their maximal effort. Before being included, they were informed about the possible risks and benefits of the project and signed their informed consent. In the first session, each subject's age, body mass, height, and regular exercise program were recorded.

Procedures

Each session started with a 10-minute warm-up on a bicycle ergometer. SDL, HE, and SKB were performed under gravitational loading conditions using free-weight equipment or under inertial loading conditions using the EPTE Inertial Concept (Ionclinics SL). EPTE Inertial Concept is an inertial device that combines a series of exit pulleys and telescopic arms, being able to implement 6 discs of different weights and radii (Figure 1). Therefore, this device allows a large number of exercises to be performed, for both upper and lower limbs, with a wide range of inertias. The first session was used to optimize and standardize the technique and, in the case of inertial loading, to become familiarized with the device. Thus, relevant measurements such as footrest distance in SKB, Roman chair height in HE, or grip and feet width in SDL were noted and used in the subsequent sessions.

In the second session, the subjects were asked to reach the maximum concentric power in each exercise. Power was calculated in both modalities as the product of velocity and force. The velocity was calculated using the MuscleLab 4020e linear encoder (Ergotest Technology AS). For gravitational loading (ie, constant mass), the force was calculated as the product of mass and acceleration. For inertial loading, the force was calculated using a force gauge that was anchored along the rope, between a pulley and the proximal end of the second rope.

All exercises began in a concentric phase, and subjects were instructed to apply maximum force. Accordingly, in sets of 6 repetitions, the subjects had between 3 and 5 attempts to achieve maximum power, with 5 minutes rest between sets.[1] The load (gravitational or inertial) was increased until a loss of the average power was noted in a set. The load applied in the series with the highest mean power was used in the subsequent session for EMG analysis.

Figure 3. Bilateral stiff-leg deadlift: gravitational (A) and inertial (B).

In the third session, MuscleLab was also used to measure EMG activation. Before sensor placement, the skin was shaved, lightly abraded, and washed with alcohol.[28] Bipolar pregelled Ag/AgCl EMG electrodes (10 mm in diameter, 15 mm interelectrode distance) (Ambu, BlueSensor N) were used. The general guidelines for electrode placement outlined by SENIAM (surface EMG for a noninvasive assessment of muscles)[17] were followed and the electrodes were secured with tape to minimize motion artifacts. Maximum voluntary isometric contraction (MVIC) took place after the warm-up and sensor placement to normalize subsequent dynamic EMG data. For GM testing, a resisted HE was requested with the subjects lying in the prone position with their legs straight.[2] Hamstrings, with subjects in prone position, were evaluated by performing knee flexor MVICs with a knee angle of approximately 45°.[2] Finally, GC was evaluated in a supine position with the hips at 45° of flexion, requesting an MVIC adduction with a resistance in the internal condyle.[20] A series of 6 repetitions of each exercise was then performed following the same order and using the load achieved in the second session.

The information from the different sensors (ie, linear encoder, force gauge, and EMG) was collected and synchronized into the Data Synchronization Unit ML6000 and processed by the MuscleLab software (MuscleLab V10.9).[32] The EMG signals were sampled at a rate of 1000 Hz. Signals were band-pass (fourth-order Butterworth filter) filtered with a cutoff frequency of 20 and 500 Hz, rectified, integrated, and converted to root-mean-square signals using a hardware circuit network (frequency

Table 1. Normalized electromyographic activity of each muscle by exercise and modality in the concentric phase[a]

		Unilateral Straight Knee Bridge	45° of Hip Extension	Stiff-Leg Deadlift	Interdevice Differences[b]
Muscle	Modality	Mean (SD)	Mean (SD)	Mean (SD)	
Biceps femoris	Gravitational	63.6 (27.0)	58.68 (21.70)	38.63 (19.06)	SKB-SDL;
	Inertial[c]	76.4 (35.7)	79.28 (28.54)	52.79 (26.11)	HE-SDL
Semitendinosus	Gravitational	53.07 (19.22)	39.18 (11.58)	37.90 (16.88)	SKB-SDL;
	Inertial[c]	63.34 (27.83)	57.16 (20.03)	39.26 (15.35)	HE-SDL
Gluteus maximus	Gravitational	35.05 (28.72)	27.21 (20.18)	34.07 (25.84)	—
	Inertial[c]	54.71 (28.01)	53.91 (28.51)	51.92 (26.70)	
Gracilis	Gravitational	50.12 (25.15)	39.22 (25.92)	53.73 (40.37)	—
	Inertial	54.83 (34.83)	49.03 (37.22)	51.12 (37.19)	

[a]All values are expressed in percent.
[b]Indicate statistical significance ($P \leq 0.05$). HE, 45° of hip extension; SDL, stiff-leg deadlift; SKB, unilateral straight knee bridge.
[c]Differences between gravitational and inertial ($P \leq 0.05$).

response 450 kHz, averaging constant 12 ms, total error ± 0.5%). For both concentric and eccentric phases, identified with the linear encoder, the filtered EMG signal was normalized to values obtained during MVIC, and these normalized EMG (nEMG) values were averaged across 5 repetitions (first repetition was discarded).

Statistical Analysis

For all analyses, SPSS was used (Version 24; IBM Corp). Participant characteristics are presented as mean (SD). Descriptive statistics were calculated for mean nEMG amplitudes of BF, ST, GM, and GC for the concentric and eccentric phases of each exercise by modality. In addition, the hamstring activation ratio was determined by dividing the average BF nEMG amplitude by the average ST nEMG amplitude (BF/ST).

For the EMG of each muscle and ratio, a 2-way repeated-measures analysis of variance was used to determine differences between modalities, the 3 exercises, and the interaction between modalities and exercises. When a significant main effect was detected, post hoc *t* tests with Bonferroni corrections were used to determine the identity of the differences. The level for significance was set at $P < 0.05$. Effect size was evaluated with η_p^2 (partial eta-squared), where $0.01 < \eta_p^2 < 0.06$ constitutes a small effect, a medium effect being $0.06 < \eta_p^2 < 0.14$, and a large effect being $\eta_p^2 > 0.14$.[10]

A prior analysis showed that 15 patients would give a power of at least 0.8 and a type I error probability of 5% for testing the hypothesis of whether there is an exercise × modality interaction. In this analysis, we deemed clinically important a mean η_p^2 effect size >0.4 among EMG measurements.

RESULTS

Muscle Differences by Phases

Table 1 shows the nEMG means for the BF, ST, GM, and GC by modality and exercise in the concentric phase. Regarding comparisons between modalities, inertial loading showed an increased nEMG as compared with gravitational loading in 3 of the 4 muscles; 15% ($P = 0.003$; $\eta_p^2 = 0.48$) greater for the BF, 9.9% ($P = 0.007$; $\eta_p^2 = 0.42$) greater for the ST, and 21.5% ($P = 0.001$; $\eta_p^2 = 0.56$) greater for the GM.

Moreover, comparisons between exercises revealed that SKB achieved a greater activation of BF, by 24.3% ($P = 0.007$), and ST, by 19.6% ($P = 0.005$) as opposed to SDL. Similarly, HE offered a higher nEMG than SDL for BF (23.3%; $P = 0.001$) and ST 9.6% ($P = 0.007$).

Table 2 shows the nEMG means for BF, ST, GM, and GC in the eccentric phase, by exercise and modality. Comparisons between modalities showed differences for GM nEMG, whereby the values for inertial training were 18.1% higher than those obtained for gravitational training ($P = 0.003$; $\eta_p^2 = 0.48$). Comparisons between exercises showed differences in the nEMG of BF and ST. For BF, the nEMG was greater in SKB and HE than in SDL (24.4%; $P = 0.003$ and 16.4%; $P = 0.002$, respectively). ST was activated 15.1% more by SKB than by SDL ($P = 0.008$). In addition, there was a significant 2-way interaction between modality and exercise in GM nEMG ($P = 0.009$, $\eta_p^2 = 0.52$), with higher nEMG values for inertial SKB with regard to the gravitational form (51.04% vs 26.31%; $P = 0.001$), as well as for inertial SDL relative to gravitational SDL (55.22% vs 32.48%; $P = 0.005$).

Table 2. Normalized electromyographic activity of each muscle by exercise and modality in the eccentric phase[a]

		Unilateral Straight Knee Bridge	45° of Hip Extension	Stiff-Leg Deadlift	Interdevice Differences[b]
Muscle	Modality	Mean (SD)	Mean (SD)	Mean (SD)	
Biceps femoris	Gravitational	55.78 (24.66)	40.45 (10.69)	28.97 (13.64)	SKB-SDL;
	Inertial	55.97 (26.03)	55.29 (23.21)	34.01 (17.38)	HE-SDL
Semitendinosus	Gravitational	43.99 (18.51)	30.11 (11.82)	31.71 (12.56)	SKB-SDL
	Inertial	48.16 (19.63)	42.84 (18.41)	30.90 (15.24)	
Gluteus maximus	Gravitational	26.31 (16.73)	26.56 (19.96)	32.48 (24.93)	—
	Inertial[c]	51.04 (23.65)	33.39 (17.23)	55.22 (39.07)	
Gracilis	Gravitational	39.42 (25.45)	38.22 (25.82)	55.36 (38.30)	—
	Inertial	34.50 (13.20)	35.33 (22.47)	47.50 (40.29)	

[a]All values are expressed in percent.
[b]Indicate statistical significance ($P \leq 0.05$). HE, 45° of hip extension; SDL, stiff-leg deadlift; SKB, unilateral straight knee bridge.
[c]Differences between gravitational and inertial ($P \leq 0.05$).

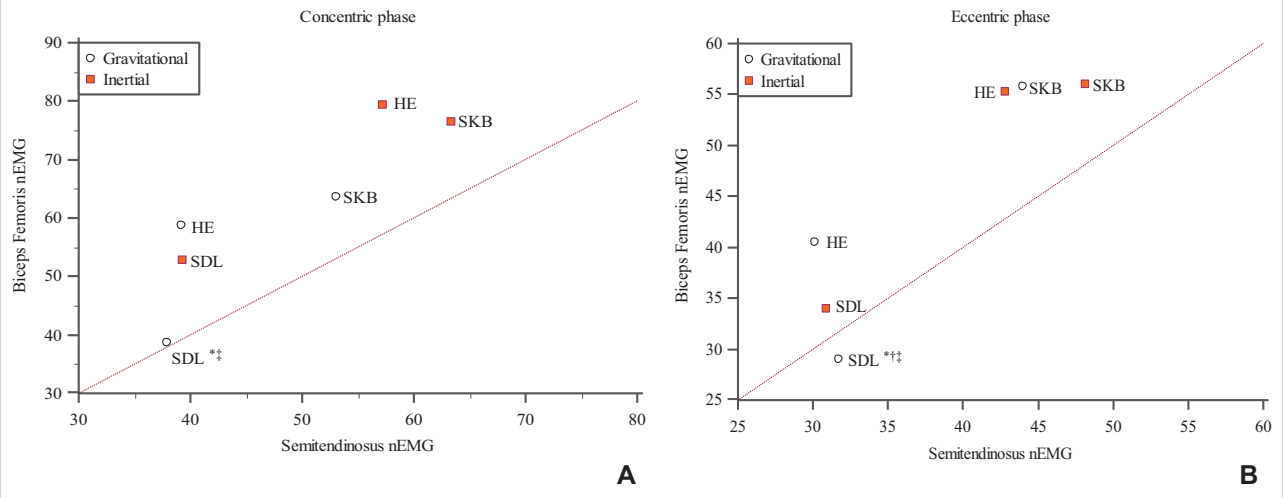

Figure 4. Biceps femoris (BF) to semitendinosus (ST) nEMG relationship for the (A) concentric and (B) eccentric phases of each exercise and modality. Exercises to the left of and above the 45° line exhibited higher levels of BF than ST nEMG. HE, 45° of hip extension; nEMG, normalized electromyography; SDL, bilateral stiff-leg deadlift; SKB, unilateral straight knee bridge. *Differences between gravitational and inertial ($P \leq 0.05$). †Significant differences compared to gravitational HE. ‡Significant differences compared to gravitational SDL ($P \leq 0.05$).

Hamstring Ratio Differences by Phases

Figure 4 shows the relationship between BF and ST nEMG by exercise and modality. All the ratios, except the gravitational SDL in the eccentric phase, presented values >1, indicating higher BF nEMG values in relation to ST.

While there were no differences between modalities, comparisons of ratios between exercises showed differences in both the concentric ($P = 0.01$, $\eta_p^2 = 0.31$) and the eccentric phase ($P = 0.001$, $\eta_p^2 = 0.51$). Specifically, in both phases, HE provided a higher ratio than SDL, the difference being 0.26 for the concentric phase (1.50 vs 1.24; $P = 0.002$) and 0.36 for the eccentric phase (1.41 vs 1.05; $P = 0.001$). In addition, there was a significant 2-way interaction between modality and exercise in eccentric ratios ($P = 0.02$, $\eta_p^2 = 0.25$), with a higher ratio for inertial SDL than for gravitational SDL (1.17% vs 0.93; $P = 0.03$).

DISCUSSION

This study examined possible differences in the EMG of individual muscles and hamstring ratios when conducting HE exercises under gravitational or inertial loading conditions. Three important findings emerged: First, inertial loading activated the hamstrings to a greater extent in the concentric phase and the GM in both phases than gravitational loading. Second, comparisons between exercises showed that SKB and HE achieved increased activation of the hamstrings in both phases as compared with SDL. Third, HE achieved a higher BF/ST ratio than SDL.

This study provides evidence that inertial loading produces a greater activation of the hamstrings only in concentric phase compared with gravitational loading. This is an unexpected finding, since inertial training is characterized by an important eccentric component, and thus, we expected that it would generate greater activation than gravitational training for the eccentric phase.[24] Even so, the differences found for the concentric phase suggest that inertial loading in HE exercises is more effective than gravitational training for hamstring activation. Furthermore, inertial training showed greater GM activation than gravitational training in both phases. Force generation is dependent on motor unit activation,[19] so high levels of nEMG are an important stimulus for improving strength and voluntary activation. Therefore, since the 3 exercises under study are outstanding within strength and conditioning programs,[4,31] our findings would suggest that performing those exercises using inertial loading might be more appropriate based on the increased muscle activation.

The nEMG values obtained in our study for gravitational exercises are partially expected.[6,23] Similarly to what was reported by Bourne et al,[6] SKB showed the highest activation levels for BF and ST in both concentric and eccentric phases, followed by HE and SDL. In contrast, while our eccentric hamstring activation levels were similar to those reported by a previous study,[6] concentric-phase nEMG of the BF and ST were moderately lower in SKB and slightly lower in HE and SDL. One possible reason for these differences is the type of subjects evaluated, since Bourne et al[6] included recreationally active athletes, while our study recruited resistance trained athletes. Hence, our subjects' wider experience and familiarization with these exercises could influence their muscle activation levels. Moreover, the GM activation levels for SDL are consistent with previous studies.[23]

To our knowledge, no previous studies have provided activation levels for inertial HE exercises.[21] Both SKB and HE were shown to be superior to SDL for concentric and eccentric hamstring activation. Although HE movements predominate in all 3 exercises, SKB and HE involve posterior heel support. Accordingly, this point where the lower limb makes its lever arm is possibly the reason why both exercises in both phases activate the hamstring bellies to a greater extent than SDL. This finding would suggest that both SKB and HE are more appropriate than SDL for inertial training of the hamstrings.

Hamstring ratios obtained showed that SKB, HE, and SDL cause greater BF activation as compared with ST activation (ratio >1). While the gravitational hamstring ratios were similar to those reported in a previous study,[6] our study provides novel results for inertial hamstring ratios. HE was shown to promote a more selective BF activation than SDL, with higher values for the BF/ST ratio. In turn, inertial SDL showed higher ratios than gravitational SDL. Exercise selectivity has been proposed as a factor with important implications for the rehabilitation of hamstring injuries by clinicians.[6] This has been argued based on the long-term eccentric inhibition of the BF after injury[26] and its influence on persistent eccentric knee flexor weakness[5] and atrophy.[29] Therefore, HE would be more suitable in connection with BF injuries, while SDL is recommended under inertial loading conditions.

Limitations

Despite these novel findings, this study was subject to limitations. First, only healthy subjects without recent injuries were evaluated. Therefore, the generalization of the patterns of selective muscle activation evidenced in our study would be limited for athletes with a history of strain injury or with another background. The second limitation was that only 3 exercises were included in our study. Although such HE exercises are selective BF exercises, other ST-predominant exercises (ie, knee flexion movements) could be included to determine consistency with gravitational ratios obtained by other authors.[6] Third, the concentric power was used as the training variable that determined the load for EMG measurement during the exercises. Although this variable was used to compare modalities under similar conditions, it could influence the differences between inertial and gravitational training for the eccentric phase.

CONCLUSION

This study demonstrates the advantages of inertial over gravitational training for concentric hamstring activation and GM activation in concentric/eccentric phases. Thus, training and rehabilitation programs that use these exercises for hamstring and GM activation should apply inertial loading conditions. On the other hand, SDL was less effective for hamstring activation than HE and SKB, with a low specificity of BF activation, especially in the gravitational form. Conversely, HE and SKB revealed high specificity for BF. Therefore, these 2 exercises should be preferentially considered over SDL in hamstring training programs.

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